



# TIPS & TRICKS

## The 'strip' command

`strip` command removes information, including the symbols (addresses of functions and variables that help in relocation) that are not required for execution from the executables. It helps in reducing the size of the file, removes information that is not required from the object file and also makes it difficult to reverse engineer the code before distribution. But debugging becomes more difficult after `strip` has been used.

```
$strip trialfile
```

Like most other Linux commands, this utility also has numerous options available for the user. The `-s` option removes all symbols.

```
$strip -s trialfile
```

The `-o` option can be used to create a new stripped file instead of the original file being modified.

```
$strip -o stripped_trialfile trialfile
```

Options are available to preserve the time stamp of the stripped file (the `-p` option), to strip a particular symbol or remove all symbols except a particular one, to get a verbose output to see the work being done when the command is executing (the `-v` option), etc. The `-R` flag can be used to remove sections, which makes reverse engineering even more difficult.

You can check the difference in the files by running the `file` command before and after the `strip` command.

—Sathyanarayanan S, [sssih@edu.in](mailto:sssih@edu.in)

## Run Java instructions on `jshell`

The Java Development Kit 9 has introduced the command `jshell`, using which you can execute the Java instructions in interpreter mode. To install this, just type the following command:

```
$ sudo apt install openjdk-9-jdk-headless
```

Then, start the `jshell` on the terminal, as follows:

```
$ jshell
```

```
-> int a = 45, b = 12, c;  
-> c = a + b  
-> c
```

—Tushar Kute, [tushar@tusharkute.com](mailto:tushar@tusharkute.com)

## Directory listing with `ls` and wildcard

Normal Linux users use `ls` for listing all the files and folders of a directory, but `ls` can be used with the wildcard characters to list matching files and folders. Such a filtered listing can be useful looking up files quickly.

For example, the following command on the terminal will only list the files and folders starting with 'a'.

```
$ ls a*
```

The output is:

```
ajit.txt astro
```

```
anaconda2:
```

```
bin  envs  lib      mkspecs  plugins  share      var  
conda-meta  etc      libexec  phrasebooks  qml      ssl  
doc  include LICENSE.txt  pkgs      sbin  
translations
```

But the above command output has one small glitch—it will also list all the files and folders (e.g., all contents of the `anaconda2` directory are listed) within the matching folders (for those folder names that start with 'a').

The solution is the `'-d'` flag of `ls`. The man page describes `'-d'` as "List directories themselves, not their contents." So the above command will become...

```
$ ls -d a*
```